

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457503.9089 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.219 +/- 0.06
Transit depth (Rp/Rs)^2	4.81 +/- 2.61 [%]
Orbital Inclination (inc)	82.93 +/- 2.11
Airmass coefficient 1 (a1)	1.09 +/- 0.13
Airmass coefficient 2 (a2)	-0.043 +/- 0.021
Scatter in the residuals of the lightcurve fit is	12.21 %
Best Comparison Star	#1 - [317, 396]
Optimal Aperture	5.34
Optimal Annulus	14.26
Transit Duration (day)	0.068 +/- 0.025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.219 ± 0.06 vs 0.1667 (deviation 0.71σ)
Duration fit vs published	0.068 d vs 0.095 d (deviation 0.88σ)
Mid-time O–C	-5.7 min
Depth SNR	3.0
Residual scatter	12.21 %

Light curve

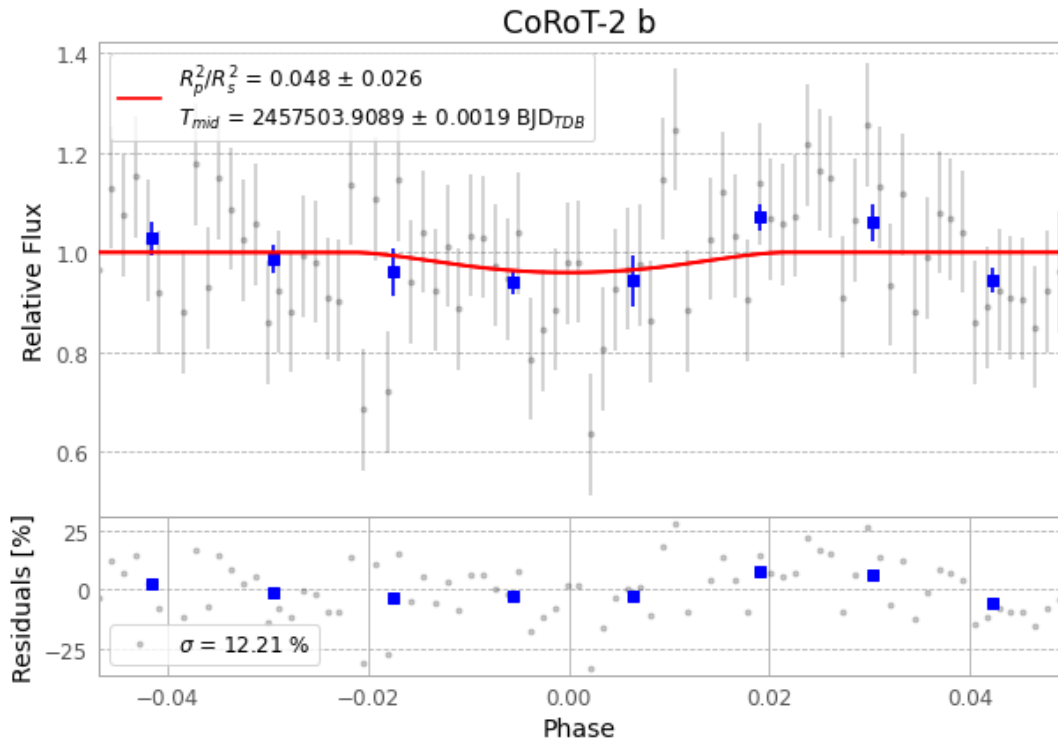


Figure 1 — FinalLightCurve_CoRoT-2 b_2016-04-25.png (SHA-256 55ace0cbeed6d1af...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [385, 394], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9f6d01b1bcdd83e1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 9131c47

Data provenance

81 FITS frames (53 MB), CoRoT-2160425074812.FITS → CoRoT-2160425114812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-160425.log (967b7505ec7b...), FinalLightCurve_CoRoT-2 b_2016-04-25.png (55ace0cbeed6...), FinalLightCurve_CoRoT-2 b_2016-04-25.csv (dbc493d33491...)

Compute resources

Wall time 115.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 06:35Z.